Aquamesh

Modular IoT Mesh-Networking Solution for Remote Water Quality Monitoring

UC San Diego
THE BASEMENT

Blackstone LaunchPad & Techstars

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Core Team:

- Alex Towfigh Founder/CEO - BS Computer Engineering
- Scott Webster Lead Mesh Engineer- BS Computer Engineering
- David Truong Lead Sensor Engineer - BS Electrical Engineering
- Rana BS Electrical Engineering
- Jay BS Mechanical Engineering

Mission: Empowering communities globally with accurate and real-time water quality data, ensuring safe water for all.

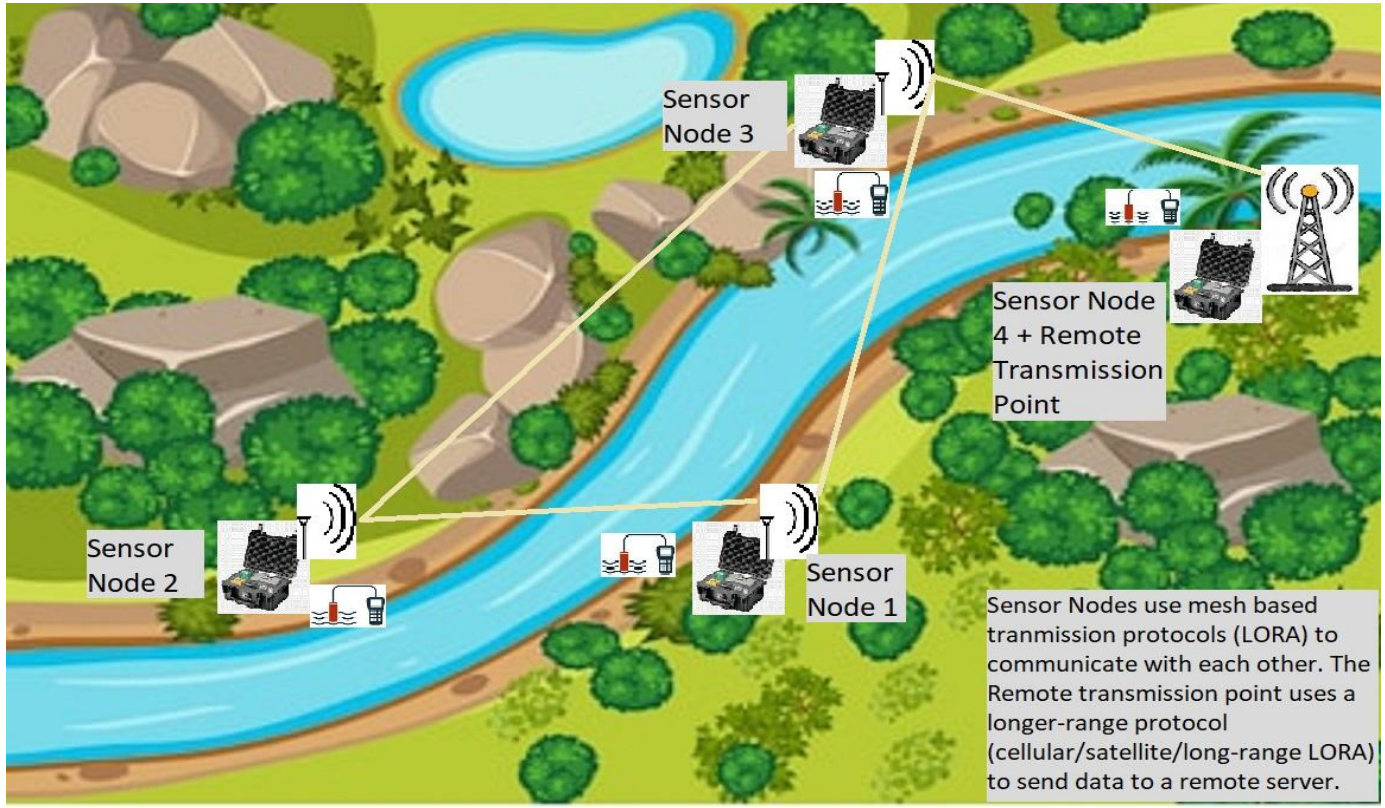
Values:

1. **Innovation: Pushing technological boundaries for global good.**
2. **Integrity: Delivering authentic, reliable water data.**
3. **Collaboration: Partnering with communities for sustainable solutions.**
4. **Empowerment: Equipping regions with tools to make informed water choices.**

Problem:

- Global Challenge: Millions worldwide lack access to clean water.
- Data Gap: Insufficient or non-existent water quality monitoring in remote or developing areas.
- Infrastructure Limitations: Heavy reliance on expensive satellite and cellular comms, limiting scalability.

Example Deployment Diagram:



Market

- Target Regions: Developing countries, remote regions, and disaster-hit areas.
- Market Size: Over 2 billion people globally lack consistent access to safe drinking water. Water Quality Monitoring Market size was valued at USD 5.13 Billion in 2021 and is projected to reach USD 9.12 Billion by 2030, growing at a CAGR of 6.6% from 2022 to 2030.
- Opportunity: Infrastructure projects, NGO partnerships, government collaborations.

Competition

- Current Solutions: Diversified and not a cohesive end-to-end solution. (In depth description in speaker notes)
- Differential Edge: Vertical integration, cost-effectiveness, modularity, real-time AI analysis.

Sales and Marketing

- Channels: Partnerships with NGOs, direct sales to governments, and collaborations with local community organizations.
- Strategies:
- Awareness Campaigns: Highlighting water quality challenges.
- Demonstration Projects/Pilots: Showcasing AquaMesh's efficacy in real-world scenarios.

Metrics & Accomplishments

- Prototype: finished working versions of mesh network and low-cost TLF sensor.
- Pilot test: planning a pilot test in the coming months with either Scripps or City of SD

Next Steps

- Continue iteration and integration
- Conduct Pilot
- Team Expansion
- Seeking more funding and grants